

gradient

Event Data Specification

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Version 2.6

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1. Introduction

This document contains the data specification of Gradient event data for association football games. Gradient events are separated into game events and possession events. Game events could represent a possession for a player on the ball, or events such as substitutions, ball out of play events, end of half/game, etc. Possession events are the events within the possession of a player, such as a pass or a cross for example. Game events can consist of multiple possession events. For example, when a player receives the ball and is challenged by a defender before passing it to a teammate, the game event consists of two possession events: a challenge and a pass. Game events and possession events are merged for your convenience. The variables on the data frame are described in the table below.

Column name	Column type	Column description
GAME_ID	int	A unique identifier for games.
COMPETITION_ID	int	A unique identifier for the competition
COMPETITION_NAME	str	Name of the competition
SEASON	str	Season in the format “year-year” or “year” (if not across two years)
GAME_EVENT_ID	int	A unique identifier for game events.
POSSESSION_EVENT_ID	int	A unique identifier for possession events.
START_TIME	float	Time the possession starts in seconds. (In reference to the start of the video, not the game)
END_TIME	float	Time the possession ends in seconds. (In reference to the start of the video, not the game)
DURATION	float	Duration of the possession in seconds.
EVENT_TIME	float	Time in seconds when the event took place. In reference to the start of the video
SEQUENCE	int	Number of the sequence. A sequence is represented by an uninterrupted possession by one team.
VIDEO_URL	str	Link to video for the specific possession
GAME_EVENTS	dict	Dictionary variable containing information regarding the game events. See section 2 .
INITIAL_TOUCH	dict	Dictionary variable containing information regarding a player's first touch. See section 3 .

POSSESSION_EVENTS	dict	Dictionary variable containing information regarding the possession events. See section 4 .
FOULS	dict	Dictionary variable containing information regarding fouls. See section 5 .
GRADES	dict	Dictionary variable containing the player grades. See section 6 .
HOME_PLAYERS	List of dicts	List of dictionary variables containing information regarding the broadcast tracking data for the players on the home team. See section 7.1
AWAY_PLAYERS	List of dicts	List of dictionary variables containing information regarding the broadcast tracking data for the players on the away team. See section 7.1
BALL	dict	Dictionary variable containing information regarding the broadcast tracking data for the ball. See section 7.2
STADIUM_METADATA	dict	Dictionary variable containing information regarding the stadium metadata. See section 8.

2. Game Event Information

As previously mentioned the game events represent a possession by a player or other game events like substitutions, or the ball going out of play. The gameEvents dict add information to the entirety of a player's possessions as well as the additional information for players coming on and off the game on a substitution for example. The Table below shows the information contained in the game event dict.

Column name (type)	Column description	Column values
GAME_EVENT_TYPE (str)	Type of game event.	1KO - first half kick off
		2KO - second half kick off
		3KO - 1st half of extra time
		4KO - 2nd half of extra time
		FOUL - Additional foul. More information on the Foul Columns section .
		END - End of half or game
		G - Ball hits the woodwork or corner flag and comes back into play
		OFF - Player comes off the pitch
		ON - Player come on the pitch
		OTB - A possession with a player on the ball
		OUT - Ball out of play

		SUB - Substitution
INITIAL_NON_EVENT (bool)	Was the possession disallowed after the fact?	True - Event disallowed
		False - Event not disallowed
START_GAME_CLOCK (int)	Time the possession starts in second in reference to the start of the game	
START_FORMATTED_GAME_CLOCK (str)	Time the possession starts in the "00:00" format.	
PERIOD (str)	The period (or half) the event took place.	1 - First Half
		2 - Second Half
		3 - First Half of Extra Time
		4 - Second Half of the Extra Time
VIDEO_MISSING (bool)	Is part of the event cut by the broadcast?	True - Part of the event is missing
		False - Entire possession is visible
HOME_TEAM	Boolean that is True if the team in possession is the home team.	
TEAM_ID (float)	A unique identifier for the team.	
PLAYER_ID (float)	Unique identifier for the player in possession.	
TOUCHES (float)	Total number of separate touches a player takes within their possession.	
TOUCHES_IN_BOX (float)	Number of touches in the opposition box a player takes within their possession.	
SETPIECE_TYPE (str)	Type of set piece being taken or type O when not a set piece	C - Corner Kick
		D - Drop Ball
		F - Free Kick
		G - Goal Kick
		K - Kickoff
		O - Open Play
		P - Penalty
		T - Throw in
EARLY_DISTRIBUTION (bool)	Is the set piece taken quickly?	True - Team tries to take the set piece quickly
		False - Team does not make an effort to take the set piece more quickly than normal.
END_TYPE (str)	Indicates the type of end event.	1 - End of First Half
		2 - End of Second Half (only used if there is extra time, otherwise it will be G for end of the game)

		F - End of the first half of the extra time
		G - End of Game
OUT_TYPE (str)	Indicates why the ball has gone out of play	A - Away Score
		H - Home Score
		T - Out of Touch
		W - Whistle
SUB_TYPE (str)	Type of substitution	H - Head Injury
		S - Standard
PLAYER_OFF_ID (float)	Unique Identifier for the player coming off the pitch	
PLAYER_OFF_TYPE (str)	Reason for the player leaving the pitch	E - Equipment e.g. having to change boots
		I - Injury
		M - Miscellaneous
		R - Red Card
PLAYER_ON_ID (float)	Unique Identifier for the player coming on the pitch	

3. Initial Touch Information

The initial touch information is stored in a dict column. Within the dict we have the following variables.

The pressure data is part of the initial touch (as well as the events). An attempt to apply pressure is recorded when a member of the opposing team is within 5 yards of the ball carrier and closing the space to them with intensity. A pressure is deemed to be successful if that opposing player closes to within 3 yards of the ball carrier, whilst continuing to close the space and affecting either the ball carrier or their options (for example cutting off pass lanes). Pressure will not be applied if the opposing team member is static or moving away from the ball carrier. Both the 5 and 3 yard radius are used as a guide and can expand or shrink depending on the intensity of the presser or direction of the pressure (smaller radius from behind). For example a player might be 4 yards from the ball carrier but at a full sprint could get awarded a successful pressure whereas a player within 3 yards and walking towards the carrier may not.

Column name (type)	Column description	Column values
INITIAL_BODY_TYPE (str)	The Body part used to take the first touch	See Appendix 1 .
INITIAL_HEIGHT_TYPE (str)	Duration of the game event (recorded in seconds).	See Appendix 2 .

INITIAL_FACING_TYPE (str)	Direction the player is facing at the point of reception.	B - Back to Goal. Not necessarily the goal itself, but general backwards direction
		G - Goal. Not necessarily the goal itself, but general forward direction.
		L - Lateral
INITIAL_TOUCH_TYPE (str)	Quality and difficulty when controlling the ball (this is used when the player retains possession, a touch bad enough to make a player lose the ball will be charted as a failed touch or heavy touch - see Touch and Ball Carry Columns for more details)	B - Hard to control, but bad
		G - Hard to control and good
		M - Miscontrol
		P - Plus. A positive touch when the ball is not necessarily hard to control. E.g. evaded a defender with a good first touch
		S - Standard

4. Possession Event Information

The possession event columns represent the general information about a specific possession event. This will be the dict that contains the highest amount of information since it captures information about all types of possession events and each has individual specific data points with some overlap. The table below will contain the metadata on the possession event and we will go into the information for each possession event type separately.

Column name (type)	Column description	Column values
POSSESSION_EVENT_TYPE(str)	Type of event.	BC - Ball Carry
		CH - Challenge
		CL - Clearance
		CR - Cross
		FO - Foul. More information on the Foul Columns section .
		IT - Initial touch (if there is another event on the initial touch the possessionEventType will be the other event instead)
		PA - Pass

		RE - Rebound
		SH - Shot
		TC - Touch. More details in section 9 .
NON_EVENT (bool)	Is the event disallowed after the fact?	True - Event disallowed
		False - Event not disallowed
EVENT_GAME_CLOCK (float)	Time in seconds when the event took place. In reference to the start of the game	
EVENT_FORMATTED_GAME_CLOCK (str)	Time when the event took place in the “00:00” format.	
BALL_HEIGHT_TYPE (str)	Height from where the ball was played.	See Appendix 2 .
BODY_TYPE (str)	Body part used to hit the ball	See Appendix 1
HIGH_POINT_TYPE (str)	Maximum height the ball reached from the point of the distribution and the next event (or the next bounce)	See Appendix 2 - Option V - half volley does not apply here.

4.1. Passes and Crosses Information.

Passes and crosses are events where the player in possession is attempting to play the ball to a teammate. We define crosses as a distribution usually from a wide area and often in the air in an attempt to create a shot opportunity in the box for a teammate. There are a few borderline cases:

- From a free kick from a position that is not so wide we can have the players set up in the box expecting for a cross, so at times we could have crosses from slightly more central positions.
- Drilling the ball across the face of the goal from a wide area will be a cross even though the ball is played low
- If instead of drilling across the face of the goal, the player in possession plays a ball on the ground or low and backwards, usually into around the height of the penalty spot or the top of the box, we would chart them as cutback passes instead of a cross.

Passes and crosses share a lot of data points such as target player, receiver player, etc. With that in mind we grouped those two events into one category. There are a few more data points that are in this group that might be used in other categories for example blockerPlayerId could also be used on shots and clearances, but all the variables that are applicable outside passes and crosses will be specified in the column description.

Column name (type)	Column description	Column values
PASSER_PLAYER_ID (float)	Unique identifier for the player passing the ball.	
PASS_TYPE (str)	Type of pass played. We chart the intention of the pass, for example, an over the top pass that was underhit and ended up creating a contest will still be type O.	B - Cutback (pass from a wide area and close to the byline pulling the ball back towards the top of the box)
		C - Creates Contest (pass played down the field trying to create an aerial duel)
		F - Flick On (Player flicks a ball that is moving forward just redirecting the ball using the momentum of the previous pass)
		H - Long Throw to Box (Throw in played deep in the box)
		O - Over the top (Ball played over a line of defense encouraging a run by the target)
		S - Standard Pass (Most common pass type, that does not fit in any of the other categories)
		T - Through ball (Ball played past a line of defense putting a player through, similar to over the top, but played at a lower height)
		W - Switch (Pass that switches the point of attack from one side of the pitch to the other)
PASS_OUTCOME_TYPE (str)	The outcome of the pass.	B - Blocked*
		C - Complete
		D - Defensive Interception*

		G - Inadvertent Shot at Own Goal
		I - Inadvertent Shot at Goal
		O - Out of Play
		S - Stoppage
CROSSER_PLAYER_ID (float)	Unique identifier for the crosser player.	
CROSS_TYPE (str)	Type of cross. We chart intention, so a miss hit cross that was intended as an outswinger cross, but ended up floating will still be type O.	D - Drilled
		F - Floated
		I - Inswinger
		O - Outswinger
		P - Placed
CROSS_ZONE (str)	Where the cross was intended to go. Since we chart intention, a cross intended to the near post that is too heavy and ends up far post, will be charted as type N.	C - Central
		N - Near Post
		S - Six Yard Box (near and far post takes precedence, so this option refers to the center of the six yard box)
		F - Far Post
CROSS_OUTCOME_TYPE (str)	The Outcome of the cross	B - Blocked*
		C - Complete
		D - Defensive Interception*
		I - Inadvertent Shot at Goal
		O - Out of Play
		S - Stoppage
TARGET_PLAYER_ID (float)	Unique identifier for the intended target of the pass or cross.	
TARGET_FACING_TYPE (str)	The direction the target is facing at the point of the pass. (Not applicable to crosses).	B - Back to Goal. Not necessarily the goal itself, but general backwards direction
		G - Goal. Not necessarily the goal itself, but general forward

		direction.
		L - Lateral
RECEIVER_PLAYER_ID (float)	Unique identifier for the teammate receiving the pass or cross. (Only applicable to completed passes)	
RECEIVER_FACING_TYPE (str)	The direction the receiver is facing at the point the pass is played. (Note that if the target is the same as the receiver, those data points will be the same.	B - Back to Goal. Not necessarily the goal itself, but general backwards direction
		G - Goal. Not necessarily the goal itself, but general forward direction.
		L - Lateral
DEFENDER_PLAYER_ID (float)	Unique identified for the defender when the outcome of the pass is "D - defensive interception"	
BLOCKER_PLAYER_ID (float)	Unique identifier for the defender when the outcome of the pass is "B - Blocked". This variable also applies to clearance and shots that are blocked.	
DEFLECTOR_PLAYER_ID (float)	Unique identifier for the player who deflects the ball. A deflection will be a lighter touch that will cause only a slight redirection. This variable also applies to shots, and if a light touch takes the ball off target or the goal is given as an own goal that is considered a block rather than a deflection on a shot.	
FAILED_BLOCKER_PLAYER_ID (float)	Unique identifier for the player who attempts a block and fails. This variable also applies to shots and clearances.	
FAILED_BLOCKER2_PLAYER_ID, FAILED_BLOCKER3_PLAYER_ID, ... (float)	If there are more than one block attempts failed, those are going to be recorded as the failedBlocker2PlayerId, failedblocker3PlayerId, and so on.	
ACCURACY_TYPE (str)	Accuracy of the pass. Exclusive for completed passes and crosses. See Appendix 3	
INCOMPLETION_REASON_TYPE (str)	The reason the pass was incomplete. See Appendix 4 for column values.	
SECOND_INCOMPLETION_REASON_TYPE(str)	We can add up to two reasons for incompleteness, so this will be the secondary reason. See Appendix 4 for column values.	
LINES_BROKEN_TYPE (str)	Line of defense broken by the pass. Also applicable for ball carries and dribbles where a player breaks a line of defense. See Appendix 5 for column values.	

*Block differs from defensive interception in that blocks are closer to the point of the distribution. If the defender is within 5 yards of the passer we classify as a block, but that definition varies with the pace of the pass, for example, on crosses or passes that are played with more pace we could have a block a little further than 5 yards, and the opposite is also true, we could have an outcome D within 5 yards on a ball that did not have much pace.

4.2. Shooting Information

A shot is any deliberate attempt to score a goal. However, if any other event results in an inadvertent shot at goal, the information for the inadvertent shot would also be found in the columns listed below.

Column name (type)	Column description	Column values
SHOOTER_PLAYER_ID (float)	Unique identifier for the player trying to score	
BODY_MOVEMENT_TYPE (str)	The direction the shooter is moving as they take the shot.	AG - Away from goal
		LA - Laterally
		ST - Static. Not moving or jumping straight up.
		TG - Moving towards goal
BALL_MOVING (bool)	Is the ball moving as the player takes the shot?	True - Ball is moving
		False - Ball is not moving.
SHOT_TYPE (str)	Type of shot attempted.	B - Bicycle (Also known as overhead kick)
		D - Diving (Player diving to meet the ball, only applicable for headers)
		F - Side Foot. (Shot hit with the medial side of the foot. However, the type is more related to the style of shot, rather than where the ball hits, a side foot shot will be a similar technique to a pass.)
		I - Sliding. (Player slides to meet the ball)

		L - Lob. Player chip or lift the ball over the goalkeeper
		O - Outside foot. (Player hits with the lateral side of the foot)
		V - Volley. (a shot hit with the ball over one ball height off the ground)
SHOT_NATURE_TYPE (str)	The nature of the shot indicates how the player tries to score	A - Placement. (Main focus on the placement of the shot, a shot that comes with power, but has the main focus on the placement will be charted as “A”.
		F - Flick. (A shot that uses the momentum of the previous event and just redirects it towards the goal. Most common example is a header from the near post just flicking to the far post)
		P - Power (Shot that focus mostly on power and less on the placement of the ball)
		T - Toe Punt (Shot that is taken with the front part of the foot)
SHOT_INITIAL_HEIGHT_TYPE (str)	The height where the goal crosses the goalline or would have crossed the goal line if not saved or blocked. This applies to events with outcome “Inadvertent Shot at Goal/Own Goal”.	‘1’ - Dividing the goal in thirds, this will be the lower part, but still off the ground.
		‘2’ - Dividing the goal in thirds this will be the middle part.
		‘3’ - Dividing the goal in thirds, this will be the top, but still more than a foot under the crossbar
		C - Cross Bar. (Ball hits the cross bar, but does not go in. If the ball were to just lightly touch the cross bar and go in it would be classified as “U -

		Narrowly under the cross bar"
		F - Fails to Reach (Ball never makes it to the goal line or would have failed to do so even if no one intervened. Not to be used if failed to reach due to a block or save)
		G - Ground
		N - Narrowly Over the Cross Bar. (Not more than a yard over the cross bar)
		O - Over. (more than a yard over the cross bar)
		U - Narrowly Under the Cross Bar. (Less than a foot under the cross bar)
SHOT_OUTCOME_TYPE (str)	The outcome of the shot. This applies to events with outcome "Inadvertent Shot at Goal/Own Goal".	B - Block on target. (Ball was going on target, but got blocked)
		C - Block off target. (Ball was going off target, and got blocked)
		F - Save off target. (Ball was going off target when it got saved)
		G - Goal.
		L - Goalline clearance. (Ball is past the goalkeeper and a defender stops it from going into the net)
		O - Off target.
		S - Save on target. (Ball was going on target and got saved).
KEEPER_PLAYER_ID (float)	Unique identifier for the player trying to the goalkeeper trying to save the shot. This applies to events with outcome "Inadvertent Shot at Goal/Own Goal".	
SAVE_HEIGHT_TYPE (str)	Height of the ball at the point of the save. This applies to events with outcome "Inadvertent Shot	Options are the same as the shotInitialHeightType above.

	at Goal/Own Goal”.	
SAVE_REBOUND_TYPE (str)	The direction the ball goes after being saved from the perspective of the shooter. This applies to events with outcome “Inadvertent Shot at Goal/Own Goal”.	CB - Crossbar
		GL - Behind the goal left
		GR - Behind the goal right
		HL - Behind the goal left and high
		HR - Behind the goal left and right
		L6 - Left six yard box
		LA - Into left area
		LO - Out of box left
		LP - Left post
		M6 - Middle six yard box
		MA - Into the middle area
		MO - Out of box middle
		OC - Over crossbar
		R6 - Right six yard box
		RA - Into right area
		RO - Out of box right
		RP - Right post
KEEPER_TOUCH_TYPE (str)	Body part with which the goal keeper touched the ball. Only used if the keeper touched the ball but it still ended up as a goal, or required a goalline clearance. See Appendix 1 for column values. This applies to events with outcome “Inadvertent Shot at Goal/Own Goal”.	
GL_CLEARER_PLAYER_ID (float)	Unique identifier for the player responsible for the goalline clearance (Only applicable for shotOutcomeType “L - Goalline clearance”. This applies to events with outcome “Inadvertent Shot at Goal/Own Goal”.	
BAD_PARRY (bool)	Does the goalkeeper have the opportunity to do better with their parry?	True - Keeper could have done better with the parry
		False - Keeper parries it as expected.

SAVEABLE (bool)	Did the keeper have a chance to save the shot that ended up in a goal or goalline clearance?	True - Goalkeeper could have saved
		False - Unsaveable shot

4.3. Clearance columns

A clearance happens when a player tries to play the ball away from danger without trying to deliberately find a teammate. We chart intention so even if the ball perfectly goes to a teammate it might still be a clearance rather than a pass. Other scenarios that are captured as clearances are:

- Trying to play the ball off an opponent to get a corner kick/throw-in/goal kick for their team.
- Deliberately putting the ball out of play or giving it back to an opposition, usually as a result of an injury on the pitch.

Column name (type)	Column description	Column values
CLEARER_PLAYER_ID (float)	Unique identifier for the player clearing the ball.	
CLEARANCE_OUTCOME_TYPE (str)	Outcome of the clearance	A - Inadvertent Shot at Goal
		B - Blocked
		D - Inadvertent Shot at own Goal
		E - Deliberate to Opposition. (Giving the ball back to the opposition, as a fair-play practice.)
		O - Out of Play
		P - Player
		S - Stoppage

		U - Deliberate to Touch. (Deliberately putting the ball out of touch, usually so a player could get medical attention)
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4.4. Challenge columns

Challenges include tackles (player in control of the ball and a challenger trying to win the ball from them), duels (two or more players competing for a loose ball), and dribbles (player in control trying to take on a defender who could attempt a tackle or not in the process). The challenge type distinguishes between those subcategories.

On tackles especially we go a little away from the industry standard. Every tackle attempt gets charted, not only if they successfully take the ball away from the carrier, and we define the winner by their performance on the challenge itself, rather than who the ball goes to after the tackle. For example, player A has the ball and player B attempts a tackle. That will be charted as a challenge. If the tackle causes player A to lose the ball player B is the winner even if the ball ends up with a player on Player A's team, unless player A was in control and intentionally allowed the ball to go to their teammate (that will be charted with a touch event for a "takeover").

The ballCarrierPlayer id and name will be also applicable on ball carries as it represents the player in control of the ball.

Column name (type)	Column description	Column values
CARRIER_PLAYER_ID (float)	Unique identifier for the player in control of the ball. NOT applicable to duels.	
DRIBBLER_PLAYER_ID (float)	Unique identifier for the player attempting a dribble.	
CHALLENGE_TYPE (str)	Type of Challenge	'5' - 50/50. This is a duel type where two players compete for a loose ball.
		A - Aerial duel. As the name suggests a duel type similar to 50-50, but with the ball coming from above.
		B - Tackle from behind. As the

		name suggests a tackle attempt where the carrier puts their body in between the ball and the challenger as the tackle is attempted.
		D - Dribble. The player tries to take on a defender in an attempt to get past them.
		G - Goalkeeper smothers ball. A duel between the goalkeeper and a line player where the ball is loose and the goalkeeper tries to capture the ball.
		H - Shielding. Similar to tackle from behind, but on a shielding challenge the carrier actively shields a defender who does not attempt a tackle
		K - Hand tackle by goalkeeper. Despite the name, it is a duel type similar to goalkeeper smothers, but the keeper tries to parry the ball rather than retain it.
		L - Slide tackle. Tackle type where the challenger slides to attempt to win the ball. Note that a player could be sliding on a dribble or 50-50, to be classed as a slide tackle it needs to be first and foremost a tackle.
		S - Shoulder to shoulder. Tackle type where the challenger tries to win the ball with physical contact initiated with the body.
		T - Standing tackle. Tackle attempt, usually from the front or side, that does not fit the other tackle types
DRIBBLE_TYPE (str)	Type of dribble attempted, as the name suggests, only applicable for dribbles.	B - Between two defenders
		I - Inside (away from the edge of the pitch)

		K - Knocks in front
		O - Outside (towards the edge of the pitch)
		T - Trick (skill move that will be described by the trickType variable)
TACKLE_ATTEMPT_TYPE (str)	Within a dribble the challenger may or may not attempt a tackle. The tackle type is captured here. Only applicable to dribbles.	D - Deliberate foul (Player does not try to play the ball)
		F - No Tackle Fake Attempt (Challenger does not attempt a tackle as a result of a fake shot/cross/pass/clearance and may have tried to go for a block instead)
		G - Defender attempts to win the ball
		T - No tackle attempt
TRICK_TYPE (str)	Type of type attempted, only applicable to dribbles with dribbleType - Trick.	C - Crujff turn. Chops and similar trick types are also captured here
		D - Drag back. Also known as pull back.
		E - Elastico. The player moves the ball to one side and then the other in one continuous move. Also known as flipflap or snake.
		F - Flick over. Also known as chip over or sombrero, etc.
		N - Nutmeg. The player attempts to put the ball through a defender's legs.
		O - Other - Trick that does not fit into another category
		R - Roulette - Also known as the Zidane, the Marseille turn or the Maradona turn/spin.
		S - Step Over

		U - Dummy - the player lets the ball run to beat an incoming defender.
CHALLENGER_PLAYER_ID (float)	Unique identifier for the primary challenger attempting a tackle, or being taken on by a dribble	
CHALLENGER2_PLAYER_ID, CHALLENGER_PLAYER3_ID, ... (float)	Unique identifier for the additional challengers. We can have multiple challengers within a challenge, which are captured in those columns, it continues on the same pattern if more are needed.	
HOME_DUEL_PLAYER_ID (float)	Unique identifier for the player on the home team primarily involved in a duel.	
AWAY_DUEL_PLAYER_ID (float)	Unique identifier for the player on the away team primarily involved in a duel.	
ADDITIONAL_DUELER_PLAYER_ID, ADDITIONAL_DUELER2_PLAYER_ID, ... (str)	Unique identifier for the additional player involved in the duel. Similar to tackles we can have multiple players in a duel, so we add columns following the same format with the more duel players	
CHALLENGE_KEEPER_PLAYER_ID (float)	Unique identifier for the goalkeeper involved in challenges of Type "G - Goalkeeper Smothers" or "H - Handtackle"	
CHALLENGE_WINNER_PLAYER_ID (float)	Unique identifier for the player who wins the challenge*	
CHALLENGE_OUTCOME_TYPE (str)	Outcome of the challenge, some outcomes are only used in dribbles and some are not used in duels.	B - Distribution disrupted, player in possession still shoots/crosses/passes/clears the ball, but that the challenger successfully disrupts the distribution
		C - Forces carrier out of play. Not applicable to duels. Ball goes out of play having touched the carrier last.
		D - Distributes ball. Player in possession passes/crosses/shoots/clears the ball right after the challenge

		F - Foul
		I - Shields in play. Not applicable to duels. The player in possession shields the ball for a teammate.
		K - Keeps ball with contact. Not applicable to duels. The player in possession retains possession even though the challenger manages to touch the ball.
		L - Rolls. Not applicable to duels. Usually on challenges from behind, the carrier has their back to the challenger and uses the contact as leverage to roll by the defender
		M - Beats man loses ball. Only applicable to dribbles. The dribbler beats the defender they are trying to take on, but loses the ball in the process.
		N - Keeps ball without beating man. Only applicable to dribbles. The dribbler keeps the ball, but fails to beat the defender they are trying to take on.
		O - Ball out of play
		P - Player. Ball goes to a player not involved in the challenge.
		R - Retain. Winner of the challenge retains possession.
		S - Shields out of play. Not applicable to duels. Carrier shields the defender allowing the ball to go out of play

4.5. Touch and Ball Carry Columns

Touches cover a variety of events that will be described on touchType, whereas ball carries captures an attempt by a player to create space, escape a pressure or break a line of defense by driving or changing directions.

Column name (type)	Column description	Column values
BALL_CARRIER_PLAYER_ID (float)	Unique identifier for the player in control of the ball.	
TOUCH_PLAYER_ID	Touches cover failing to control the ball, deliberate handball, and failed distributions. This is the unique identifier for situations not including failed distributions (covered specifically bellow) this player ID also includes missed touches.	
CARRY_TYPE (str)	Type of action that describes the ball carry.	B - Line break. If a player breaks a line by driving or changing direction, we would pick those options instead.
		C - Change of direction
		D - Drive with intent
BALL_CARRY_OUTCOME (str)	Outcome of the ball carry attempted.	C - Leads into Challenge. Would need to be a duel (challenger where the player is no longer in control). If the ball carrier is still in a position of control on the challenge the outcome will be Retain.
		L - Ball Loss
		R - Retains
		S - Stoppage
CARRY_DEFENDER_PLAYER_ID (float)	Unique identifier for the player trying to defend the carry.	
CARRY_INTENT (str)	What the ball carriers intends to accomplish on the carry	B - Break line, if the intention is to create space or escape a pressure and the player also breaks a line, we would default to the other options.

		C - Create space. A bit more proactive trying to create space around a defender.
		E - Escape Pressure. A bit more reactive trying to escape the pressure by the defender
CARRY_SUCCESSFUL (bool)	Did the player successfully accomplish what they intended to do?	True - Player successfully accomplish what their carryIntent was
		False - Player failed to accomplish their carryIntent
TOUCH_TYPE (str)	Touch encompasses a variety of events which are defined by the type of touch.	C - Failed cross. The player attempts to cross and misses the ball completely, the ball then hits them on a different body part (standing foot for example).
		D - Deliberate handball. A line player deliberately handles the ball or the keeper handles the ball from a back pass. (Just making their body bigger, or other handball would go as rebounds)
		F - Failed trap. The player fails to control the ball.
		G - Failed Catch. Goalkeeper attempts to catch the ball and fails.
		H - Heavy touch. The player controls the ball, but the touch is a bit too heavy. (Taking a heavy touch while changing direction or driving will be a ball carry with outcome L - ball loss instead)
		L - Failed clearance. The player attempts to clear and misses the ball completely, the ball then hits them on a different body part (standing foot for example).
		P - Failed pass. The player attempts to pass and misses the ball completely, the ball then hits them

		on a different body part (standing foot for example).
		S - Failed shot. The player attempts to shoot and misses the ball completely, the ball then hits them on a different body part (standing foot for example).
		T - Take over. The player deliberately allows their teammate to take over the ball without passing to them.
TOUCH_OUTCOME_TYPE (str)	The outcome of a touch event.	A - Inadvertent shot at Goal
		C - Challenge. Similar to ball carries, it would need to lead into a duel (challenge where neither player is in a position of control)
		D - Inadvertent shot at own goal
		O - Out of play
		P - Player
		R - Retain

4.6. Rebound Columns

Rebounds are collected on a touch where a player has no intention to control or distribute. The most common examples of rebounds are saves where the goalkeeper is not trying to catch just parry, blocks on shots/passes/crosses/clearances, and any inadvertent touch. We would also capture a rebound out of a challenge if it leads into an accidental goal or own goal.

Column name (type)	Column description	Column values
REBOUNDER_PLAYER_ID (float)	Unique identifier for the player the ball hits.	
ORIGINATE_TYPE (str)	Where the ball comes from leading to the rebound.	C - Corner flag
		M - Miscellaneous
		P - Player

		W - Woodwork
REBOUND_OUTCOME_TYPE (str)	Outcome of the rebound	A - Inadvertent shot at goal
		D - Inadvertent shot at own goal
		P - Player
		T - Out of touch

4.7. Additional Event Information

The information below are columns that will apply to all or most of the possession events, which are: missed touch information, pressure (on the event rather than on the initial touch, see a more detailed explanation for pressure in [section 3](#)), opportunity types, and many others. Those data points will be further explained below.

Column name (type)	Column description	Column values
MISSED_TOUCH_TYPE (str)	Similar to touch events, missed touches do not get a game event as the player fails to make contact with the ball, instead a missed touch will be captured within the previous event.	C - Missed Cross (the player tries to cross and completely misses the ball)
		D - Dummy (the player intentionally lets the ball run for a teammate, usually trying to attract a defender in the process)
		I - Missed Interception (the player misses the ball in an attempt to intercept the ball)
		L - Missed Clearance (the player tries to clear and completely misses the ball)
		M - Missed Touch (the player tries to control the ball and completely misses)
		O - Failed Shot (the player tries to shoot and completely misses the ball)
		P - Failed Pass (the player tries to pass and completely misses the ball)
		S - Slip (the player misses the ball as a result of a slip)
MISSED_TOUCH_PLAYER_ID (float)	Unique identifier for the player who missed the ball.	
INITIAL_PRESSURE_PLAYER_TYPE (str)	Type of pressure being applied at the first touch of a possession.	A - Attempted Pressure
		L - Passing Lane Pressure
		N - No Pressure

		P - Player Pressured
INITIAL_PRESSURE_PLAYER_ID (float)	Unique identifier for the player applying the pressure on the first touch	
MP_PRESSURE_PLAYER_TYPE (str)	Type of pressure being applied. MP_PRESSURE is collected when a pressure occurs mid-possession, not in a particular event	A - Attempted Pressure
		L - Passing Lane Pressure
		N - No Pressure
		P - Player Pressured
MP_PRESSURE_PLAYER_ID (float)	Unique identifier for the player applying the pressure mid-possession.	
MP_PRESSURE_PLAYER_TIME (float)	Time where the mid-possession pressure occurred.	
PRESSURE_TYPE (str)	Type of pressure being applied. Same options as the initial touch pressure, but referring to a pressure on the event instead.	A - Attempted Pressure
		L - Passing Lane Pressure
		N - No Pressure
		P - Player Pressured
PRESSURE_PLAYER_ID (float)	Unique identifier for the player who pressured the player in possession.	
OPPORTUNITY_TYPE (str)	Type of opportunity an event generates for their team or the opposing team.	C - Chance created (receiver should score)
		D - Dangerous position (receiver in position to create a chance)
		H - Half chance (receiver has a decent chance to score)
		N - Negative chance created (player has created a chance or half chance for the opposing team)
		P - Negative dangerous position (player has created a dangerous position for the opposing team)
BETTER_OPTION_TYPE (str)	We would chart a better option if a player missed an opportunity to do something better with their possession. This captures the type of better Option	B - Ball Carry. The player should have attempted a carry or a dribble. Also used if a player should have left the ball to another player
		C - Cross. The player should have attempted a cross
		H - Hold. The player should have held on to the ball instead of distributing it.

		L - Clearance. The player should have cleared the ball.
		O - Continue. The player should have continued a forward run rather than stopping or changing directions.
		P - Pass. The player should have passed the ball
		S - Shot. The player should have taken a shot.
BETTER_OPTION_TIME (float)	Time in seconds in reference to the start of the video (not the game) where the opportunity presented itself. For example, if we think a player should have passed early we would have better option pass with an earlier timestamp.	
BETTER_OPTION_PLAYER_ID (float)	This is the unique identifier for the target of the pass, cross or takeover (ball carry). One scenario which is different is if there is a dummy and the better option would be for the target not to dummy it, in that case this data point refers to the player who attempted the dummy	
CREATES_SPACE (bool)	Did the player in possession attempt to create space prior to a distribution (even if unsuccessful)?	True - There was an attempt to create space
		False - No attempt to create space.
FAILED_CLEARER_PLAYER_ID (float)	Unique identified for the player trying to clear the ball and failing	
FAILED_CROSSER_PLAYER_ID (float)	Unique identified for the player trying to cross the ball and failing	
FAILED_PASSER_PLAYER_ID (float)	Unique identified for the player trying to pass the ball and failing	
FAILED_SHOOTER_PLAYER_ID (float)	Unique identified for the player trying to shoot the ball and failing	

4.8. Sub-grade Players

A grade given will refer to a specific player, for example, a passing grade would be referring to the passingPlayerId, which is collected in the [passing section](#). However, we have the subgrades which address the players away from the ball. (e.g positionPlayerId). Those data points are solely collected for the purpose of grading the players, and those columns are addressed below. All of those data points allow for more than one player to

be collected per possession and they will have the following format: positionPlayerId, positionPlayer2Id, positionPlayer3Id, etc.

Column name (type)	Column description
POSITION_PLAYER_ID, POSITION2_PLAYER_ID, ... (float)	Unique identifier for a player who will receive a positional grade. We can have multiple positional grades in a possession and they will follow the same format, increasing the number.
CLOSING_DOWN_PLAYER_ID, CLOSING_DOWN2_PLAYER_ID, ... (float)	Unique identifier for a player who will receive a closing down grade. We can have multiple positional grades in a possession and they will follow the same format, increasing the number.
DAB_PLAYER_ID, DAB2_PLAYER_ID, ... (float)	Unique identifier for a player who will receive a DAB (duel away from the ball) grade. We can have multiple positional grades in a possession and they will follow the same format, increasing the number.
MOVEMENT_PLAYER_ID, MOVEMENT2_PLAYER_ID, ... (float)	Unique identifier for a player who will receive a movement grade. We can have multiple positional grades in a possession and they will follow the same format, increasing the number.
DISCIPLINE_PLAYER_ID, DISCIPLINE2_PLAYER_ID, ... (float)	Unique identifier for a player who will receive a discipline grade. We can have multiple positional grades in a possession and they will follow the same format, increasing the number.

5. Foul Information

Foul information is stored into a dict called “fouls”. We capture the information for fouls that are given, advantages, and fouls that our data collectors deem to have been missed by the referee. All “onField” columns represent the on field decision and the “final” columns the final decision, note that the values will be the same if there is no VAR.

We can have multiple infringements in an event which are added separately (for example an on the ball foul, then a yellow card for dissent added in two separate fouls). When that is the case, the main foul will be added on the possession event where it happened and the additional fouls will be added as a separate row after the next OUT events. It will have gameId “FOUL”, and possessionEventId “FO” even though technically it is not a separate event.

Column name (type)	Column description	Column values
ON_FIELD_CULPRIT_PLAYER_ID (float)	Unique identifier for the player who committed the foul, as per the referee's initial decision.	
FINAL_CULPRIT_PLAYER_ID (float)	Unique identifier for the player who committed the foul, as per the referee's final decision, usually the same unless var is used for a card that is given to the wrong player.	
VICTIM_PLAYER_ID (float)	Unique identifier for the player who is the victim of the foul.	
FOUL_TYPE (str)	Foul type defines if a foul is given, missed or an advantage is played.	A - Advantage
		I - Infringement
		M - Missed infringement
OFFENSE_TYPE (str)	Unique identifier for the offense type.	D - Dissent. We incorporate verbal and physical violent acts towards the referee and/or other players all into dissent.
		F - Off the ball
		H - Handball
		N - On the ball
		O - Offside
		T - Technical. Incorporates fouled throw-ins, delaying the game, direct shot out of an indirect free kick, etc.
ON_FIELD_FOUL_OUTCOME_TYPE (str)	The outcome of the foul, similarly to the other data points the on field and final outcome will be the same unless VAR is used.	V - Dive
		F - No foul. Foul was not awarded, only used for missed infringements and when there is VAR interference.
		N - No warning. Foul is given and no warning or card is awarded.
		R - Red card

FINAL_FOUL_OUTCOME_TYPE (str)		S - Second yellow card
		W - Warning. This is only selected if there is a clear shot on the broadcast showing the referee giving the culprit a warning, if we see them talking, but unsure it is a warning, we chart as no warning.
		Y - Yellow card
VAR (bool)	Was VAR used on the play? VAR is technically used in the background in many scenarios, like offsides, goals, possible penalties, so we only select "True" when the referee needs to go to the screen or the initial decision is overturned.	True - An on-field review (OFR) is recommended, or the initial decision is overturned by VAR.
		False - VAR is not used or var just confirms the referee's initial decision without recommending an on-field review (OFR).
VAR_REASON_TYPE (str)	The reason VAR was used.	I - Missed infringement. (Infringement was missed at first and VAR used to potentially award the infringement).
		O - Overturn Infringement. (The Infringement was initially awarded, and VAR used to potentially overturn the decision).
CORRECT_DECISION (bool)	Was the final decision correct?	True - Correct decision by the referee and/or VAR.
		False - Incorrect decision by the referee and/or VAR.

6. Grades

The grading system evaluates every player on every event during a game. Each player is given a grade of -2 to +2 in 0.5 increments on a given event with 0 being the expected

grade. The zero grade is important as most events feature many players doing their job at an expected level.

A grade given will refer to a specific player, for example, a passing grade would be referring to the passingPlayerId, which is collected in the [passing section](#). However, we have the subgrades which address the players away from the ball, which are addressed in this section. (e.g positionPlayerId).

Column name (all float)	Column description
ADDITIONAL_DUELER_GRADE, ADDITIONAL_DUELER2_GRADE, ...	As seen on the challenge columns , we can have multiple additional duelers and each column will reflect the grade corresponding to their respective additionalDuelerPlayerId.
AWAY_DUEL_GRADE	Grade given to the primary player from the away team involved in a duel, which would correspond to the awayDuelPlayerId.
BALL_CARRIER_GRADE	Grade given corresponding to the ballCarrierPlayerId.
BLOCKER_GRADE	Grade given to the player blocking a distribution, corresponding to the blockerPlayerId.
CARRIER_GRADE	Grade given to the player to the player in control of the ball on a tackle. Corresponding to the carrierPlayerId.
CARRY_DEFENDER_GRADE	Grade given to the player defending a carry, which corresponds to the carryDefenderPlayerId.
CHALLENGE_KEEPER_GRADE	Grade given to a goalkeeper on a goalkeeper smothers or hand tackle, which corresponds to challengeKeeperPlayerId.
CHALLENGER_GRADE, CHALLENGER2_GRADE, ...	As seen on the challenge columns , we can have multiple challengers on a tackle or dribble as well, and the challenger grades will correspond to the equivalent challengerPlayerId.
CLEARER_GRADE	Grade given to the player clearing the ball, which corresponds to the clearerPlayerId.
CLOSING_DOWN_GRADE, CLOSING_DOWN2_GRADE.	Grade given to the player for their attempt (or lack thereof) to close down on the player in possession. As seen on the sub-grade players section , we can have multiple closing down players, and each grade will correspond to their respective closingDownPlayerID.

CROSSER_GRADE	Grade given to the crosser, which corresponds to the <code>crossePlayerId</code>
DAB_GRADE, DAB2_GRADE, ...	DAB stands for defending away from the ball, but it encompasses a bit more than that. The DAB grade can be awarded for blocking off a defender and opening space for another player on a corner for example, or other situations in that nature. As seen on the sub-grade players section , this grade allows for multiple DABs per possession and they will correspond to their respective <code>dabPlayerId</code>
DEFENDER_GRADE	Grade given to the defender that gets the ball on an incomplete pass or cross, which corresponds to the <code>defenderPlayerId</code> .
DEFLECTOR_GRADE	Grade given to the player who deflects a distribution, which corresponds to the <code>deflectorPlayerId</code> .
DISCIPLINE_GRADE, DISCIPLINE2_GRADE, ...	Grade (usually downgrade) given for discipline, most common examples are verbal or physical violent conduct, trying to delay the game, etc. As seen on the sub-grade players section , it is possible to have multiple discipline grades per possession event, so they will correspond to their equivalent <code>disciplinePlayerId</code> .
DRIBBLE_GRADE	Grade given to the player trying to dribble past a defender, this will correspond to the <code>dribblerPlayerId</code> .
FAILED_BLOCKER_GRADE, FAILED_BLOCKER2_GRADE, ...	Grade given to the player attempting to block a distribution and failing. As seen on the Passes and Crosses Columns , we can have multiple per possession event, so the grades will correspond to their equivalent <code>failedBlockerGrade</code> .
FAILED_CLEARER_GRADE	Grade given to a player who failed to clear the ball as a result of a touch type “Failed Clearance” or a missed touch type “Missed clearance” (See Touch and Ball Carry Columns or Additional Event Information). The grade will correspond to the <code>failedClearerPlayerId</code> .
FAILED_CROSSER_GRADE	Grade given to a player who failed to cross the ball as a result of a touch type “Failed Cross” or a missed touch type “Missed cross” (See Touch and Ball Carry Columns or Additional Event Information). The grade will correspond to the <code>failedCrosserPlayerId</code> .
FAILED_PASSER_GRADE	Grade given to a player who failed to pass the ball as a result of a touch type “Failed Pass” or a missed touch type “Missed pass” (See Touch and Ball Carry Columns or Additional Event Information).

	The grade will correspond to the failedPasserPlayerId.
FAILED_SHOOTER_GRADE	Grade given for a player who failed to shoot the ball as a result of a touch type “Failed Shot” or a missed touch type “Missed shot” (See Touch and Ball Carry Columns or Additional Event Information). The grade will correspond to the failedShooterPlayerId.
GL_CLEARER_GRADE	Grade given to the player responsible for a goalline clearance, which will correspond to the glClearerPlayerId.
HOME_DUEL_GRADE	Grade given to the primary player from the home team involved in a duel, which would correspond to the homeDuelPlayerId
KEEPER_GRADE	Grade given to a keeper for their effort to save a shot or inadvertent shot, which would correspond to the keeperPlayerId.
MOVEMENT_GRADE, MOVEMENT2_GRADE, ...	Grade given for an off the ball movement by a teammate. As seen on the sub-grade players section , it is possible to have multiple movement grades per possession event, so they will correspond to their equivalent movementPlayerId.
PASSER_GRADE	Grade given to the player passing the ball, which will correspond to the passerPlayerId.
POSITION_GRADE, POSITION2_GRADE,...	Grade given to a defensive player for their positioning. As seen on the sub-grade players section , it is possible to have multiple position grades per possession event, so they will correspond to their equivalent positionPlayerId.
RECEIVER_GRADE	A receiver grade will be the same as a movement grade, but specific to the recipient of a pass or cross. The grade will correspond to the receiverPlayerId.
SHOOTER_GRADE	Grade given to the player attempting to score, which will correspond to the shooterPlayerId.
TARGET_GRADE	A target grade will be the same as a movement grade, but specific to the target of a pass. The grade will correspond to the targetPlayerId.
TOUCH_GRADE	A touch grade will be given to the player on a touch or missed touch event where the player was trying to control the ball or intercept a distribution. This grade will correspond to the touchPlayerId.

7. Location Data

The location data comes from broadcast tracking and is stored in three different columns: homePlayers, awayPlayers and ball.

7.1. Home and Away Player Tracking Information

Those two columns contain a list of dicts which stores information about all the home/away players on the pitch. They will both contain the same columns listed below.

Column name (type)	Column description	Column values
PLAYER_ID (int)	Unique identifier for the player in question.	
JERSEY_NUM (int)	Number of the Jersey of the player in question.	
POSITION_GROUP_TYPE (str)	Position of the player in question.	
CONFIDENCE (str)	Degree of confidence in the information given in the tracking data.	LOW
		MEDIUM
		HIGH
VISIBILITY (str)	Whether the player was visible or estimated in the broadcast.	ESTIMATED
		VISIBLE
X (float)	Player's x coordinate	
Y (float)	Player's y coordinate	

7.2. Ball Tracking Information

The Ball column contains a dict with the information about the location of the ball.

Column name	Column description	Column values
VISIBILITY (str)	This data point describes whether the location of the ball is estimated or visible.	ESTIMATED
		VISIBLE
X (float)	Ball's x coordinate	
Y (float)	Ball's y coordinate	
Z (float)	Ball's z coordinate	

8. Stadium Metadata

This column contains information about the stadium and the pitch dimensions and the side the home team started the game.

Column name	Column description
STADIUM_ID	Unique identifier for the stadium
stadiumName	Name of the Stadium
PITCH_LENGTH	Length of the pitch in meters
PITCH_WIDTH	Width of the pitch in meters
HOME_TEAM_START_LEFT	Boolean that is True if the home team started on the left side (attacking right)
HOME_TEAM_START_LEFT_EXTRA_TIME	Boolean that is True if the home team started on the left side (attacking right) in the first half of the extra time.
TEAM_ATTACKING_DIRECTION	Direction that the team in possession is attacking. Will be represented as 'R' for right, 'L' for left, and empty if no team is in possession (ball out of play events for example)

Appendix 1 - Body part Options

The table below shows the body part type values that we capture.

Body type	Description
2H	Two hands - mostly throw-ins or goalkeeper distributions with two hands, most other goalkeeper events will have CA, PA and PU.
BA	Back
BO	Bottom
CA	Two hand catch
CH	Chest
HE	Head
LA	Left arm
LB	Left back heel
LC	Left shoulder
LF	Left foot
LH	Left hand
LK	Left knee
LS	Left shin
LT	Left thigh
PA	Two hand palm
PU	Two hand punch
RA	Right arm
RB	Right back heel
RC	Right shoulder
RF	Right foot
RH	Right hand

RK	Right knee
RS	Right shin
RT	Right thigh
VM	Video missing

Appendix 2 - Height Options

The table below shows the options for heightType as well as highPointType on the event level and on the initial touch.

Height is measure by an average standing player, so it's is not variable by the body position of a specific player, for example, a player jumping and getting the ball with their chest could be "A - above head" and a diving headed could be "L - off the ground, but below waist".

Note that despite the name "A - Above Head" this is captured more as "head and above", meaning a non-jumping header would get height type A (unless the player is lowering their body to meet the ball"

Height type	Description
A	Above head
G	On the ground
H	Between waist and head
L	Off the ground but below waist
M	Video missing
V	Half volley - a ball that is hit as it comes up after bouncing. The difference between half volley and off ground but below waist is that for a half volley the ball will still be rising when hit. NOT APPLICABLE TO HIGH POINT.

Appendix 3 - Accuracy Type

The table below shows the accuracy type values that are available in any pass or cross that have an outcome - “C - Complete”.

Accuracy Type	Description
A	Away from defender - The passer/crosser deliberately puts the ball away from an incoming defender, putting the ball in a position where their teammate can more a better chance to receive the ball more cleanly
C	Checks Movement - The pass/cross forces the target to abruptly stop their run to receive the ball
E	Leads into challenge - The opposite of away from defender, not an inaccurate pass per se, but puts the ball into a position where the target would have a harder time dealing with an incoming pressure or challenge. This will rarely be used on Crosses as there is some expectation that there will be a challenge when playing a cross with defenders in the box.
H	Heavy - The direction of the pass/cross is fine, but comes with a pace that makes the ball harder for the receiver to control/shoot or forces a small adjustment by the target.
I	In Stride - A pass or cross played into a moving target who does not have to adjust their feet or their speed to receive the ball
L	Light - Direction of the pass or cross is fine, but the pass lacks pace and causes the target to have to wait for the ball, at times inviting pressure/challenge.
P	Precise - A pass or cross played through or into a tight window
R	Redirects - A pass or cross that forces a target to change direction to collect/shoot the ball. Note that redirects would take precedence over the other accuracy types
S	Standard - Most common type, will be the default if a pass or cross does not match another category.

Appendix 4 - Incompletion Reason Type

The table below shows the incompletion reason type values that are available in any pass or cross that does not have an outcome - “C - Complete”.

Incompletion reason type	Description
BH	Behind
BL	Blocked
CA	Caught
CO	Defensive Contact
DC	Defensive Challenge
DF	Deflected
DI	Defender Interception
FO	Foul
HI	High
HO	Hit Official
IF	In Front
LB	Receiver Lets Ball Run
MC	Miscommunication
MH	Miss Hit
PS	Passer Slipped
RB	Receiver Didn't Come Back To Ball
RF	Receiver Slipped
RM	Receiver Misses Ball
RS	Receiver Stopped
RW	Referee In Way
SP	Speculative
UH	Underhit

VM	Video Missing
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Appendix 5 - Lines Broken Type

A line break is captured when a pass, a carry or a dribble takes the ball through a line of defense (not captured if passing around or over a line of defense).

Lines must be made up of 2 or more players & should be built from the back moving forward. There must always be a defensive line, there can only be 1 defensive & 1 midfield line, but there can be multiple attacking lines. So if a teams defensive shape is 4-1-4-1 for example, we would have just 2 lines, a defensive line of 4 & a midfield line of 4, however if a teams shape was 4-2-2-2, we would have 4 lines, a defensive line of 4, a midfield line of 2 & 2 separate attacking lines of 2.

Lines Broken Type	Description
A	Attack
AD	Attack and defense (bypassing the midfield)
AM	Attack and midfield
AMD	Attack, midfield and defense
D	Defense
M	Midfield
MD	Midfield and defense